

The Test of Auditory-Vocal Affect (TAVA) dataset

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Abstract—In this paper we present the TAVA, a novel dataset for advancing research in Speech Emotion Recognition (SER) by disentangling paralinguistic and linguistic information in affective speech signals. The dataset includes 352 audio recordings of emotionally expressive English speech, each paired with a corresponding transformed electroglottographic (tEGG) version—a signal designed to preserve affective cues while systematically suppressing phonetic content. In addition, we provide over 120,000 crowd-sourced ratings of valence, arousal, and dominance for both the original and transformed signals. These ratings support fine-grained comparisons of affect perception across modalities. Building on prior work showing that tEGG signals can effectively isolate vocal affect, this dataset offers a unique resource for evaluating sensitivity to vocal affect in clinical populations with language or communication difficulties, as well as in studies aimed at dissociating linguistic and affective processing in individuals without these impairments. By contributing a phoneme-reduced, affect-rich signal representation to the SER community, we aim to enable more robust modelling of vocal affect and broaden the applicability of SER systems to diverse user populations.

1. INTRODUCTION

The human voice expresses emotion in ways that are more foundational—and often more immediately felt—than the words we speak and their explicit meaning. Traditional approaches to speech emotion recognition (SER) typically rely on full speech signals, which blend linguistic and paralinguistic content [1]. However, there is growing recognition of the need to disentangle vocal affect from language—particularly for potential applications in clinical research, where individuals with autism, aphasia, or language impairments may struggle with interpreting or expressing emotion through speech. This ability to perceive affect in vocalizations is a fundamental feature of communication in humans and other animals. We possess dedicated neural mechanisms that automatically attribute emotional meaning to voice-like stimuli even in the absence of linguistic content [2], [3]. This perceptual sensitivity to affect is critical for social functioning: its refinement predicts social fluency [4], while its impairment is associated with conditions such as autism, schizophrenia, and major depressive disorder [5]–[7]. Isolating paralinguistic affect from phonetic content enables the assessment of sensitivity to a speaker’s ‘tone of voice’ without confounds related to language processing. This work builds on a signal transformation method based on simultaneous recordings of speech audio and electroglottography (EGG) [8]. This approach suppresses phonetic content while preserving basic affective cues—such as pitch, intensity, and voice quality—that are central to emotional expression. The transformed EGG (tEGG) signals were shown to convey affective content similar to their original speech counterparts, in a sample of speech-tEGG stimulus pairs.

In this paper, we build on this work by releasing the TAVA dataset—a new publicly available resource containing paired speech and tEGG recordings annotated with crowd-sourced ratings. By systematically removing phonetic information while preserving basic emotional tone, this dataset enables the development of SER models that are robust to linguistic variation and suitable for language-independent or clinical applications. Our contributions are as follows: (1) TAVA dataset: A curated corpus of 352 English speech audio clips and their corresponding transformed electroglottographic (tEGG) signals. The dataset spans 22 speakers (half female), each uttering 16 affective targets. (2) Emotion ratings: Over 120,000 crowd-sourced ratings of valence, arousal, and dominance, providing a dense mapping of emotional perception. (3) SER benchmark tasks: Baseline models for both regression (predicting valence, arousal, and dominance) and classification (identifying discrete affects) on the basis of acoustic features. (4) A tool for language-independent SER: By removing phonetic content while preserving paralinguistic cues, TAVA provides a resource for developing and evaluating SER models that generalize across linguistic boundaries—an asset for both clinical and cross-cultural applications.

2. RELATED WORK

SER is the computational task of identifying emotion based on how something is said, rather than what is said. The field gained traction with Picard’s Affective Computing [9], which emphasized emotional intelligence as central to the field of human-computer interaction. Since then, a wide range of SER datasets have been developed [1], [10]–[12], with an online catalog^a maintaining an updated repository. Most SER datasets are solely comprised of speech, confounding emotional expression with semantic content, and making it impossible to separate the contribution of paralinguistic and linguistic features.

Emotions are often conveyed by the voice independently of language, for example, in affect bursts (e.g., laughter, cries, screams). There is also evidence that responses to paralinguistic information are primary to linguistic information and evoke stronger, earlier neural responses than speech prosody [13], suggesting that the brain prioritizes non-verbal affective cues. To address language-related confounds and emphasize paralinguistic features, electroglottographic signals have been explored as an alternative modality. EGG captures vocal fold activity without articulatory or phonetic content. For example, Pravena et al. [14] demonstrated that combining EGG-derived features with speech can improve emotion classification, underscoring EGG’s potential for emotion recognition [15]. tEGG allows for gating

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^a<https://superkogito.github.io/SER-datasets/>

analyses and comparison between longer-form vocalizations that include temporal and prosodic variation [16], [17].

Together, these findings point to a core challenge: with the exception of affect bursts, existing SER datasets are based on speech audio in which affective and linguistic cues are inexorably intertwined. This issue is particularly acute in clinical contexts involving speech and/or language impairments (e.g., autism, aphasia, apraxia), where SER systems may benefit from focused sensitivity to paralinguistic aspects of vocal affect. Moreover, emotion-labeled non-linguistic vocal affect datasets, like that offered here, may support broader machine learning applications beyond speech, serving as training resources for affect recognition on music and other sound sources [18]–[21].

3. METHODOLOGY

3.1. Data collection

The TAVA dataset builds on prior work describing a pipeline for generating tEGG signals from speech and EGG [8]. In 2020, 44 university-affiliated “voice actors” recorded speech targeting 16 affective states—two per octant in a valence–arousal–dominance emotion model. Targets were conveyed to speakers via emotion word lists (e.g., “joyful, playful”), cartoon figures, written scenarios, and coordinates in the emotion model. Each actor read a unique 4–9 syllable sentence, designed for phonological balance and emotional neutrality (e.g., “my spoon is on the table”). Poor EGG signal quality led to exclusion of two speakers. The 22 most expressive (12 male, 10 female, 2 nonbinary; mean age 26, SD 9, range 19–49) were selected by DLB, all native North American English speakers. Recordings were made in a sound-attenuating chamber at Stanford’s Center for Computer Research in Music and Acoustics with a DPA 466 microphone and Zoom H4n PRO. EGG was recorded using a Glottal Enterprises EG2-PCX2 system (44.1 kHz; 20 Hz low-frequency cutoff) and captured via Adobe Audition (48 kHz). Electrodes were placed on either side of the larynx and secured with an elastic strap. Sound intensity was measured using an NTi XL2 sound level meter (dBA, fast weighting, 10 Hz). A clapperboard ensured cross-modal alignment. Sessions began with a vocal warm-up and EGG calibration. Participants stood while vocalizing 16 targets in fixed order, repeating each 3–10 times until both actor and experimenter were satisfied with expression quality.

Following recording, signals were synchronized and a single high-quality rendition was selected per target per speaker. EGG recordings were denoised using Adobe Audition’s DeNoise effect, and high-pass filtered when necessary to remove artifacts. To preserve intensity cues, speech audio was amplitude-scaled based on original sound levels (36–89dB @ 1m). To prevent inaudibility and clipping, the 53dB range was linearly compressed by 60% to 32dB, preserving relative differences between targets. Final amplitudes were normalized to –28 LUFS in MATLAB (EBU R-128 standard). tEGG signals were generated per [8] by modulating the EGG with the speech envelope and a learned vocal tract filter, preserving prosody while suppressing phonetic content (see Fig. 1). tEGG amplitudes were matched to corresponding speech audio.

3.2. Ratings

Affect ratings were collected online via CloudResearch^b, which recruits vetted Mechanical Turk participants with a history of attentiveness. Eligibility required native English speakers located in the U.S. After providing informed consent and basic demographics, participants adjusted their device volume using a noise stimulus

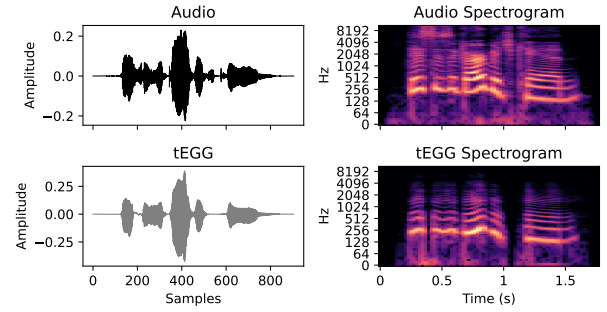


Fig. 1: Visualization of transformation for an example of audio.

matched to the average spectrum across all speech recordings, then completed a Huggins pitch screen to ensure their use of stereo headphones [22].

Following instructions and practice trials (without feedback), participants rated subsets of the TAVA stimuli in randomized order, interleaved with validated control stimuli from prior studies [23]–[25]. Some stimuli were repeated to compute intra-rater reliability via the intraclass correlation coefficient (ICC, absolute agreement, mixed effects, single rater). Ratings were made using three visual analogue scales—valence (“negative” to “positive”), arousal (“low-energy” to “high-energy”), and dominance (“submissive” to “dominant”)—with a cartoon manikin dynamically reflecting slider positions (based on [26]). Ratings were collected over six phases (June 2022 and September 2023). Raters were excluded based on low reliability on repeated items (average $ICC < 0.5$ across valence, arousal, and dominance), failure to appropriately rate control stimuli, or failure to pass embedded attention checks (10% excluded overall).

We collected affect ratings for both speech and tEGG versions across all emotional dimensions. Each stimulus in the dataset corresponds to one of 16 target emotions performed by 22 trained voice actors, with a total of 704 unique stimuli (352 speech, 352 tEGG). All audios amount to 11.4 minutes of audio including speech and tEGG. Data from a total of 1,532 unique participants was collected, (age $\mu = 41$, $\sigma = 12$, range 18–85; 45% female, 55% male, <1% nonbinary; 14% no college, 22% some college, 64% bachelors degree or higher. On average, each participant rated 78.2 stimuli (range: 30 – 352). The average number of ratings per stimulus was 260.5 (range: 58 – 941). In total, 120,312 affect ratings were recorded—86,564 for natural speech and 33,748 for tEGG-transformed speech. The TAVA dataset—along with the annotations, analysis scripts and raw EGG data—is made publicly available to ensure reproducibility of the results.^c

3.3. Benchmark

To evaluate the encoding of affective information in the speech and tEGG, we extracted emotionally-relevant features from audio using the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [18], which has been extensively employed in automatic vocal affect analysis tasks, including paralinguistic and clinical speech research. The eGeMAPS acoustic feature set has 88 parameters that relate to frequency (e.g., pitch, formants), energy (e.g., loudness), spectrum (e.g., spectral slope), and cepstrum (e.g., spectral flux and mel-frequency cepstral coefficients). Unlike large-scale brute-force feature sets, eGeMAPS offers a compact, theoretically grounded selection of acoustic parameters, chosen based on their (a) sensitivity to emotionally-relevant changes in the physiological of vocal production, (b) empirical support from previous studies, and (c) ease of automatic

^b<https://www.cloudresearch.com>

^c<https://github.com/juansgomez87/tava-data>

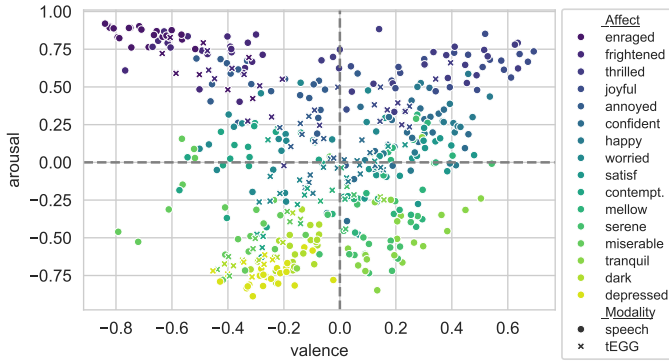


Fig. 2: Mean arousal and valence annotation for each rated speech/tEGG stimulus, ordered by mean arousal.

extraction. For each TAVA stimulus, we extracted “bag-of-frames” features per audio—specifically the mean and coefficient of variation of each parameter across individual audio fragments—using the openSMILE toolkit^d. The resulting acoustic features were used as input for two predictive tasks: categorical emotion classification and dimensional affect regression. For emotion classification, we trained a logistic regression model to distinguish among the 16 affect target categories. For dimensional affect prediction, we employed Lasso regression (linear regression with L1 regularization) to predict the mean ratings of valence, arousal, and dominance. These benchmarks serve as a baseline for assessing how well standard acoustic features capture ratings of affect in the speech and tEGG modalities.

4. RESULTS

4.1. Annotation analysis

Inter-rater reliability. To assess the consistency of affective ratings across participants, we computed inter-rater reliability using both the ICC (one-way random, single measures) and Krippendorff’s α across the three affective dimensions: valence, arousal, and dominance. ICC measures how much rating variance reflects item differences versus noise, while α measures agreement beyond chance, even with missing data. We calculate the ICC using a linear mixed-effects model to assess rating consistency across raters, particularly for sparse or unbalanced data. Overall, reliability was moderate for speech stimuli, but somewhat lower for tEGG stimuli. For speech stimuli, which included 86,564 unique ratings, ICC values were 0.74 for arousal, 0.53 for valence, and 0.49 for dominance. In contrast, for tEGG stimuli, which consisted of 33,748 unique ratings, ICC values decreased to 0.58 for arousal, 0.29 for valence, and 0.35 for dominance. These results were mirrored by Krippendorff’s α scores: for speech, the values were 0.73 for arousal, 0.53 for valence, and 0.49 for dominance, while for tEGG the values were 0.56, 0.26, and 0.34, respectively. As anticipated, valence showed lower reliability than arousal, reflecting its higher subjective variability. Despite this, the reliability for speech stimuli—measured by both ICC and α —remains moderate, whereas it is notably reduced for tEGG stimuli [27], [28]. We observed a strong positive correlation between mean arousal and mean dominance ratings across all stimuli, with $r(460) = 0.64, p < .001$ (see Fig. 2).

Speech vs. tEGG rating similarity. To evaluate the extent to which affective content is preserved in the transformation from speech to tEGG, we compared individual affective ratings across

the three dimensions—valence, arousal, and dominance—for matched speech–tEGG stimulus pairs ($n = 18,199$ ratings from 514 individuals rating 80 stimulus pairs). Pearson correlation coefficients indicated moderate linear associations between the two modalities: arousal, $r = 0.64$; valence, $r = 0.44$; and dominance, $r = 0.49$. To determine whether ratings differed significantly between the modalities, we conducted paired-sample t-tests for each affective dimension. A Bonferroni correction was applied to control for multiple comparisons, resulting in an adjusted significance threshold of $\alpha = 0.0167$. To evaluate differences in perceived emotion between the speech and tEGG conditions, we conducted paired t-tests on ratings of valence, arousal, and dominance, along with effect size estimates (Cohen’s d). For valence, the absolute values of ratings were greater in the speech condition ($M_{diff} = 0.11, t[18198] = 30.53, p < .001, d = 0.23$). Similarly, for arousal ($M_{diff} = 0.17, t[18198] = 50.82, p < .001, d = 0.38$) and for dominance ($M_{diff} = 0.12, t[18198] = 34.90, p < .001, d = 0.26$). These results indicate consistent but small differences in affective ratings across modalities, with speech eliciting slightly greater emotional intensity than tEGG. These findings suggest that while affective impressions are preserved to a substantial degree in the tEGG modality, systematic shifts in perceived affect emerge when phonetic content is suppressed. We find that speech ratings tended to be more extreme, while tEGG were more conservative. As this analysis only pertains to a subset of the data, further analysis is needed to support generalization. That said, this analysis is thorough in that 15% of ratings of each affect type were included. Figure 3 shows the distribution of all ratings grouped by the 16 target affects. For arousal, differences emerged for *enraged*, *frightened*, *worried*, *contemptuous*, *dark*, and *depressed*. For valence, significant differences were observed for *enraged*, *frightened*, *thrilled*, *annoyed*, *satisfied*, and *depressed*. For dominance, *satisfied*, *contemptuous*, *miserable*, *dark*, and *depressed* showed significant shifts. These results suggest that some types of vocal affect are more susceptible to phonetic suppression than others. For example, affects associated with highest and lowest arousal (e.g., *enraged*, *frightened*, and *depressed*) seem to especially sensitive to tEGG transformation, where this is less true for some mid-arousal affects (e.g., *confident*, *satisfied*, and *mellow*).

4.2. Benchmark results

We evaluated how well standard acoustic features capture affect by testing model performance on categorical and dimensional ratings using stratified 5-fold cross-validation.

Target affect classification. The speech-based model achieved a classification accuracy of 25.9% (*macro* – $F1 = 0.26$), whereas the accuracy of the tEGG-based model dropped to 20.7% (*macro* – $F1 = 0.21$). McNemar’s test confirmed that both classifiers performed significantly better than the chance baseline of 6.2% ($p < 0.0001$ in both cases), suggesting that affective information is present in both modalities, even though the tEGG performance was considerably worse. Class-wise performance differed substantially across modalities: (1) emotions such as *frightened*, *enraged*, and *thrilled* were among the best recognized in both speech and tEGG, although speech consistently yielded higher F1-scores; (2) the speech model showed stronger performance for *dark*, *worried*, and *contemptuous*, while *depressed* was better recognized in tEGG; (3) several positive emotions like *satisfied*, *mellow*, and *happy* showed weak performance across both modalities. Confusion matrices (see Fig. 4) highlight several consistent and modality-specific patterns of misclassification. In the speech-based classifier, common confusions occur between semantically similar categories such as *confident* and *mellow/satisfied*; *depressed* and *miserable*; *happy* and *satisfied*; as well as *mellow* and *serene*. In

^d<https://audeering.github.io/opensmile-python/>

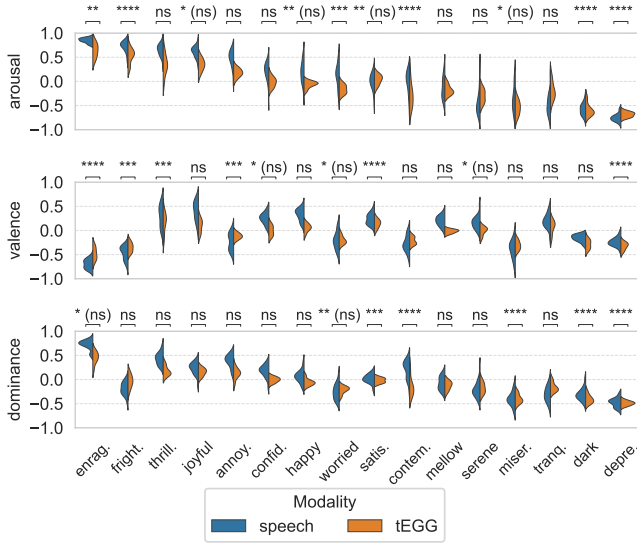


Fig. 3: Distributions of ratings for paired speech/tEGG stimuli in each affect category, ordered by mean arousal. Significance tested using independent-sample t-tests with Holm-Bonferroni correction. $p_{adj} < 0.05^*$, $<0.01^{**}$, $<0.001^{***}$, $<0.0001^{****}$, ns=not significant.

contrast, the tEGG-based classifier exhibits greater confusion between pairs like *annoyed* and *confident*, *confident* and *satisfied*, *happy* and *mellow/satisfied*, and *joyful* and *thrilled*. Notably, both classifiers struggle to distinguish between *enraged* and *frightened*, and between *serene* and *tranquil*. Interestingly, suppressing phonetic content of acoustic-based classifications results in similar confusions between speech and tEGG.

Dimensional regression. We applied LASSO regression (with optimal $\alpha = 0.01$ via grid search) to predict mean arousal, valence, and dominance ratings from speech and tEGG features. LASSO, suitable for high-dimensional data, performs feature selection by shrinking irrelevant coefficients to zero. Whereas a baseline would simply predict the sample mean as the output, the speech-based LASSO regression model achieved an R^2 of 0.89 for arousal ($MSE = 0.03$, $MAE = 0.14$), 0.29 for valence ($MSE = 0.089$, $MAE = 0.24$), and 0.58 for dominance ($MSE = 0.05$, $MAE = 0.18$). Although predictive performance explained considerable variance, t-tests comparing errors between LASSO models and baselines did not indicate significant smaller errors for LASSO models ($p > 0.8$ in all cases). This suggests insufficient data to outperform the sample mean (352 speech instances). For the tEGG-based model, results were generally more modest, with R^2 values of 0.74 for arousal ($MSE = 0.05$, $MAE = 0.17$), 0.16 for valence ($MSE = 0.04$, $MAE = 0.16$), and 0.47 for dominance ($MSE = 0.04$, $MAE = 0.14$). As with speech, however, no significant advantage over the baseline was observed in any case ($p > 0.3$)—presumably also due to insufficient data (108 tEGG instances). Although the LASSO models captured meaningful variance—especially for arousal—other acoustic features could improve the predictive signal for affective ratings. The modest R^2 values—particularly for valence—highlight both the challenge of small-sample affect modeling and the opportunity for richer, data-efficient approaches (e.g., transfer learning or data augmentation).

5. CONCLUSIONS

We introduced TAVA, a dataset for speech emotion recognition (SER) combining aligned speech and tEGG signals, over 120,000 affective ratings, and benchmark tasks for classification and regression.

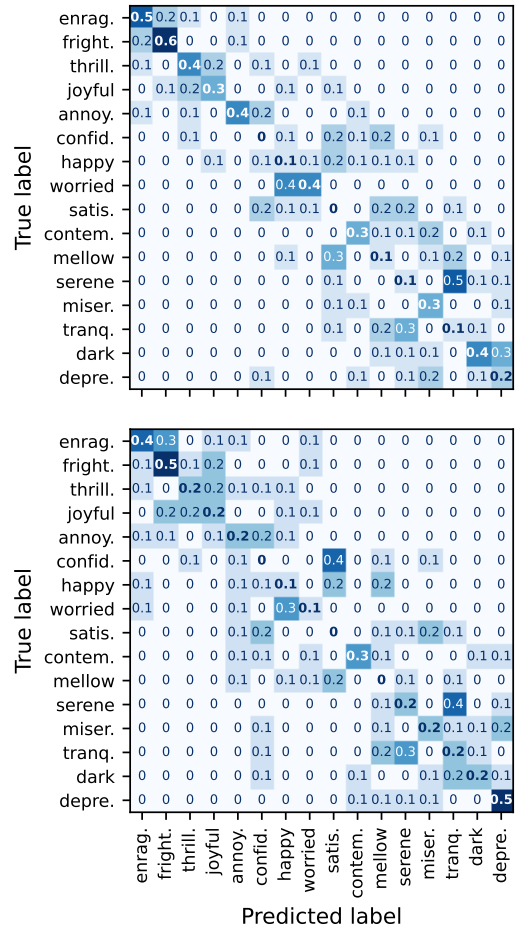


Fig. 4: Confusion matrices comparing emotion classification performance across modalities (*top*: speech, *bottom*: tEGG).

TAVA enables phoneme-free SER modeling, supporting language-independent applications. This may also exclude elements of emotional expression, limiting personalized SER. We find that speech ratings tended to be more extreme, while tEGG were more conservative. tEGG ratings showed lower inter-rater agreement, raising questions about their suitability as ground truth. Experiments with classification showed that both modalities encode affective information beyond chance, with speech outperforming tEGG overall (~ 5 percentage points F1-score)—especially for emotions like *annoyed*, *dark*, *worried*, and *thrilled*—while tEGG showed relative strength for *depressed*. In dimensional regression, arousal emerged as the most reliably predicted affective dimension in both modalities, reaching $R^2 = 0.89$ for speech and $R^2 = 0.74$ for tEGG—as seen in [1], [11], [18]. Valence and dominance were less predictable, especially in the tEGG condition, possibly due to a heavier dependence on linguistic cues during rating or imperfect preservation of paralinguistic cues in the tEGG signal. Because tEGG suppresses phonetic content, raters may struggle to assess emotion dimensions like valence and dominance—leading to greater uncertainty, lower agreement (see Sec. 4.1), and reduced algorithmic prediction accuracy for these variables. These findings validate TAVA for benchmarking SER models and highlight tEGG’s promise as a language-independent, phoneme-free signal for affective computing, paving the way for improved affect-preserving, more advanced algorithms.

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